Final Year Project Summary

*The Project summary limited to **One (1) page**. Please submit in PDF format.*

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Matric Number: BI23110344
Program: UH6481005 Data Science

Project Title: **A Multi-Modal Framework for Quantifying Narrator Reliability in 'Ilm ar-Rijal using Semantic Sentiment Analysis and Graph Neural Networks**

Supervisor: **Dr. Khalifa Chekima**

Project Type: Research

Problem Statement:

Traditional *'Ilm ar-Rijal* (narrator evaluation) is a manual, labour-intensive process requiring scholars to cross-reference thousands of qualitative biographical entries to assess the authenticity of a *Hadith isnad* (chain of transmission). While recent advancements in Natural Language Processing (NLP) have improved narrator name extraction and entity disambiguation, there is a critical lack of computational frameworks that can quantify narrator reliability. Specifically, current models fail to integrate the semantic sentiment of *Jarh wa Ta'dil* (scholarly praise or criticism), a field in hadith sciences, found in classical texts with the structural features of the transmission network.

Objective:

1. To design and implement an Arabic Named Entity Recognition (NER) model to isolate narrator names, lineages, and biographical labels, followed by an Entity Linking mechanism to resolve name variants and disambiguate narrators with similar names.
2. To develop a Natural Language Processing (NLP) classification model that maps qualitative scholarly judgments found in *Jarh wa Ta'dil* texts to a standardized, quantitative reliability index (0.0 - 1.0).
3. To construct a Knowledge Graph representing teacher-student relationships and utilise Graph Neural Networks or centrality measures to evaluate the structural integrity and temporal consistency of transmission chains.
4. To evaluate the framework's accuracy by comparing its automated reliability scores against established scholarly Hadith gradings for a selected benchmark of Hadith isnads.

Scope:

1. Data:
   - Digital text corpus of Taqrib al-Tahdhib and Tahdhib al-Kamal via OpenITI/Shamela.
   - Scale: ~1,500 biographical entries (narrator records) and ~1,000 validated Hadith *isnads* (chains) for testing.
2. Experiment Input and Output:
   - Independent Variables: Textual features, structural features, and temporal data.
   - Dependent Variables: Predicted narrator reliability score (continuous 0.0–1.0) and final chain authenticity classification (categorical: Sahih, Hasan, Dha'if).
   - Control Variables: Benchmark dataset of pre-graded Hadith from Al-Kutub al-Sittah.
3. Experiment Input and Output:
   - Input: Raw Arabic text strings of a hadith *isnad* and corresponding biographical entries.
   - Output: At narrator level, individual reliability scores and resolved identities; at chain level, a composite probability score and a predicted authenticity grade for the entire hadith.

Supervisor Stamp of approval

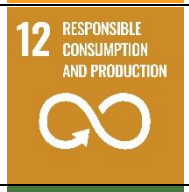
Date:

Remarks (for Review Committee use)

SUSTAINABLE DEVELOPMENT GOALS

https://sdgs.un.org/goals

Your project should aim to fulfil one of the 17 Sustainable Development Goals (SDG). Please map your project to one of the relevant SDG by ticking the relevant box.

SDG	Description	/	SDG	Description	/
	End poverty in all its forms everywhere			Reduce inequality within and among countries	
	End hunger, achieve food security and improved nutrition and promote sustainable agriculture			Make cities and human settlements inclusive, safe, resilient and sustainable	
	Ensure healthy lives and promote well-being for all at all ages			Ensure sustainable consumption and production patterns	
	Ensure inclusive and equitable quality education and promote lifelong learning opportunities for all	/		Take urgent action to combat climate change and its impacts	
	Achieve gender equality and empower all women and girls			Conserve and sustainably use the oceans, seas and marine resources for sustainable development	
	Ensure availability and sustainable management of water and sanitation for all			Protect, restore and promote sustainable use of terrestrial ecosystems, sustainably manage forests, combat desertification, and halt and reverse land degradation and halt biodiversity loss	
	Ensure access to affordable, reliable, sustainable and modern energy for all			Promote peaceful and inclusive societies for sustainable development, provide access to justice for all and build effective, accountable and inclusive institutions at all levels	
	Promote sustained, inclusive and sustainable economic growth, full and productive employment and decent work for all			Strengthen the means of implementation and revitalize the Global Partnership for Sustainable Development	
	Build resilient infrastructure, promote inclusive and sustainable industrialization and foster innovation				